

SKILL 2

1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char buffer[256];
```

```
    while (1) {
```

```
        printf("myshell> "); // Display prompt
```

```
        if (fgets(buffer, sizeof(buffer), stdin) == NULL) break; // Read input
```

```
        buffer[strcspn(buffer, "\n")] = 0; // Remove newline
```

```
        if (strcmp(buffer, "exit") == 0 || strcmp(buffer, "quit") == 0) {
```

```
            printf("Exiting shell...\n");
```

```
            break; // Exit condition
```

```
        }
```

```
        printf("You typed: %s\n", buffer); // Echo back
```

```
    }
```

```
    return 0;
```

```
}
```

```
bash: cd: too many arguments
aasritha@DESKTOP-4B676UM:~$ mkdir OSSP_skill
aasritha@DESKTOP-4B676UM:~$ cd OSSP_skill
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ nano loop.c
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ gcc loop.c -o loop
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ ./loop
myshell> hello
You typed: hello
myshell> hi
You typed: hi
myshell> exit
Exiting shell...
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ |
```

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main() {
```

```
    char buffer[256];
```

```
    int index = 0;
```

```
    int ch;
```

```
    printf("myshell> ");
```

```
    while (1) {
```

```
        ch = getchar();
```

```
        // Enter key
```

```
        if (ch == '\n') {
```

```
            buffer[index] = '\0'; // End string
```

```
            if (strcmp(buffer, "exit") == 0 || strcmp(buffer, "quit") == 0) {
```

```
                printf("\nExiting shell...\n");
```

```
                break;
```

```
            }
```

```
            printf("\nYou typed: %s\n", buffer);
```

```
            index = 0; // Reset buffer
```

```
    printf("myshell> ");
}

// Backspace
else if (ch == 127 || ch == '\b') {
    if (index > 0) {
        index--;
        printf("\b \b"); // Erase visually
    }
}

// Normal characters
else {
    if (index < sizeof(buffer) - 1) {
        buffer[index++] = ch;
        putchar(ch); // Echo character
    }
}

return 0;
}
```

```
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ nano space.c
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ gcc space.c -o space
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ ./space
myshell> hello
hello
You typed: hello
myshell> hel
hel
You typed: hel
myshell> hello
hello
You typed: hello
myshell> hi
hi
You typed: hi
myshell> exit
exit
Exiting shell...
aasritha@DESKTOP-4B676UM:~/OSSP_skill$ |
```